Natural Language Processing

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Indice

1 Introduction

Language is a structured system of conventional signs used by agents to represent, express and communicate meanings.

1.1 How did language emerge?

Two question must be distinguished:

- **The evolutionary origin of the human capacity for language:** how biological, cognitive and social evolution made language possibile.
- **The historical development of particular language:** how language changes, differentiate, interact and give rise to language families

The origin of language itself is not directly documented. Spoken and signed languages existed long before writing, so their early history must be reconstructured indirectly using evidence from:

- human biology and cognition
- archaeology
- comparative linguistic
- animal comunication

1.2 What is language?

Language is a structured system of conventional signs used by agents to represent, express and communicate meanings.

- Spoken sound
- Signed gesture
- Written symbols
- Tactile signs, such as Braille
- Haptic signs, conveyed through touch or movement

1.3 Who does use language?

- **Human agents**
- **Non-human animal agents**
  - some non-human animals use structured communication system, although their classification as languages is disputed
- **Artificial agents**
  - artificial agents process and generate linguistic expressions, while the nature of their understanding remains an open theoretical question.

1.4 Level of linguistic representation

Level	Principal units	Central question
Signal	Sounds, gestures, written or tactile signs	How is language physically conveyed?
Phonological or graphemic	Phonemes, syllables, graphemes	How are elementary signs organized and distinguished?
Morphological	Morphemes and word forms	How are meaningful components combined into words?
Lexical	Words, lemmas and senses	What words exist, and what can they mean?
Syntactic	Phrases, clauses and sentences	How are words combined into structured expressions?
Semantic	Concepts, entities, predicates, arguments and propositions	What does the expression mean?
Discourse	Sentences, utterances and discourse relations	How does an expression connect with the surrounding text or conversation?
Pragmatic	Speakers, hearers, intentions and context	What does an agent want to obtain by using the expression in this situation?

1.5 Typology based on writing system

Languages should be distinguished based on their features. We may start by their writing system to step onto four further basic features:

- Morphology
- Syntax
- Semantics
- Pragmatics

1.6 IPA

The international phonetic alphabet (IPA) is a standardized system for representing the sounds of spoken language. Ideally, each symbol identifies a particular speech sound independently of the ordinary spelling conventions of a laguage.

We can present three languages as a progression in the relationship between **written form and pronounciation**. This is not a progression from a *simple language* to a *complex language*, it concerns the complexity of their orthographic systems and form-sound correspondences.

1.7 Ambiguity

Ambiguity occurs when the same linguistic expression permits two or more distinct analyses or interpretations.

2 Language as Data

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2.1 What can be used to describe a language?

- Lexical resources
- Grammatical resources
- Phonological and writing resources
- Empirical language resources
- Documentantion resources

Resource

RICOPIARE TABELLE

2.2 What is a Corpus?

A corpus is not merely a large collection of texts. Its scientific value depends on what containes, how its components are distributed and which population of linguistic events it is intended to represent.

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3 Linguistic units

3.1 Units of language

- Documents, paragraphs and sentences
- Token and types
- World forms and lemmas
- Morphemes and subword unit
- Vocabulary and out-of-vocabolary items
- Linguistic units as modelling decision

3.2 Document, paragraph, sentece

- A **document** is a complete textual object treated as a unit within a dataset, also may be an article, book, email, review, post, transcript or webpage
- A **paragraph** is an intermediate textual unit grouping senteces that contribute to a locally coherent topic or functions
- A **sentence** is grammatically organized linguistic unit conventionally marked in marking by punctuation

$$\text{document} \supset \text{paragraph} \supset \text{sentences}$$

3.3 Word form and lemmas

- A **word forms** is the observable form in which a word occurs in a text
- A **lemma** is the conventioanl base form used to group related inflected word forms
- Different word forms may therefore correspond to the same lemma.
- Word forms preserve grammatical information such as tense, number, person and case
- Lemmas abstract away from inflection and identify the underlying lexical unit
- **Lemmatization** is the process of assigning the appropriate lemma to a word form

3.4 Morphemes and subword unit

A **morpheme** is the smallest linguistic unit conventionally associated with a meaning or grammatical function, a word may contain one or more morphemes:

$$\text{cats} \quad \text{cat} + \text{-s}$$

$$\text{unhappiness} \rightarrow \text{un-} + \text{happy} + \text{-ness}$$

- **Free morphemes** can occur independently: `cat`, `happy`.
- **Bound morphemes** occur only as parts of words: `un-`, `-s`, `-ness`
- Morphemes may express lexical meaning or grammatical information
- A **subword unit** is a unit smaller then a word used by a computional model.
- Subword units may be learned statistically from data and do not necerssarily correspond to linguistic morphemes.

3.5 Vocabulary and out-of vocabulary items

4 Introduction to ML measures

4.1 From predictions to outcomes

To evaluate a system, compare the excpected correct answer produced by the system.